

Jamie Seoh

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EDUCATION

University of Waterloo

Bachelor of Software Engineering

Sep 2025 – Present

4.0/4.0 (92.73%) GPA

- Logistics Organizer @Hack the North, Sponsorship Director @Ignition Hacks, Event Coordinator @UWKSA
- Previous President @Hack the Ridge, President @IRHS Korean Club, Conductor/Section Leader @IRHS

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript & JavaScript, HTML & CSS, SQL, Java, R

Frameworks and Libraries: PyTorch, NumPy, React.js, React Native, Next.js, Svelte, FastAPI, Flask, Tailwind CSS

Technologies: Docker, GCP, Azure, AWS, Bash, MongoDB, Supabase, Git/GitHub, VS Code, Figma

PROJECTS

Nebula AI 🍷 🍏 | *React Native, FastAPI, PostgreSQL (Supabase), Docker, GCP*

- Developed a mobile app that turns handwritten notes into a searchable, **LaTeX**-formatted AI knowledge base.
- Enabled semantic search over **1,536-dim embeddings**, cutting retrieval time by **40%** with IVFFlat indexing.
- Implemented an **OCR+AI pipeline** with Gemini/OpenAI, enabling handwritten text extraction and summaries.
- Designed multi-tenant data access using Supabase Auth, JWTs, and RLS policies across 5 relational tables.
- Containerized backend with **Docker**, deployed to **Google Cloud Run** with automated CI/CD pipelines.

Quota — DeltaHacks12 First Place 🍷 | *TypeScript, Next.js, React Flow, Gemini, LangChain, MongoDB*

- Built VS Code extension with **TypeScript** for inline cost annotations, heatmaps, and one-click optimizations.
- Indexed codebases **<3s** for large codebases with **10,000+** lines using **AST parsing** to detect inefficiencies.
- Delivered real-time API cost insights via regex and usage caching, cutting API spend by up to **40%**.
- Created **React Flow** sandbox for budget-aware system planning with **RAG Gemini** chatbot by LangChain.

Personal CRM 🍷 | *Next.js, FastAPI, PostgreSQL, TailwindCSS*

- Built a robust full-stack CRM to centralize and prioritize personal & professional relationships at scale.
- Architected backend with **20+ REST** endpoints, enabling pagination & dynamic fetching for growing datasets.
- Improved system reliability through caching layers and rate limiting, reducing server load by **30%**.
- Designed a **matrix-based algorithm** to rank contacts, enabling users to focus on high-value relationships.

Melodie.ai 🍷 | *PyTorch, NumPy, Python, Music21*

- Developed an **LSTM**-based neural network with **PyTorch** to generate folk melodies, reaching coherent sequences.
- Preprocessed MIDI data by standardizing keys and normalizing sequences for simpler, faster training.
- Designed **probabilistic sampling system** with temp. control and nucleus sampling methods, enabling variation.

EXPERIENCE

Software Developer

UW Orbital Satellite Mission Design Team | React, Figma, FastAPI

Sep 2025 – Nov 2025

Waterloo, Ontario

- Built Mission Control dashboards in **React** to monitor satellite telemetry and uplink/downlink sessions.
- Implemented **REST APIs** to serve async requests to MCC, standardizing data contracts between subsystems.
- Authored unit and integration tests, achieving **95% UI coverage** with **vitest** for Mission Control Center.

Front-End Developer

SolveXchange | HTML, CSS, JavaScript, Flask, SQLAlchemy

Sep 2023 – Jun 2024

Online

- Designed and shipped responsive frontend interfaces serving **100+** active users for a non-profit platform.
- Built secure authentication and data workflows with **Flask**, **SQLAlchemy**, and **RESTful APIs**.

Web Development Intern

PNPT Co., Ltd. | Figma, HTML, CSS, JavaScript, React

Jul 2023 – Aug 2023

Gangnam District, Seoul

- Collaborated on early-stage website optimization for a startup, enhancing usability across 3 pages.
- Prototyped 2 responsive UIs using **Figma & React**, accelerating design iterations and stakeholder feedback.
- Debugged and resolved cross-browser compatibility issues, ensuring consistent UI rendering across browsers.

PUBLICATIONS

Seoh, J. (2025). *Analyzing Literary Factors: A Big Data Approach.* STEM Fellowship Journal.
Published academic paper: "Literary Factors Analysis" (DOI #: 10.17975/sfj-2025-001).